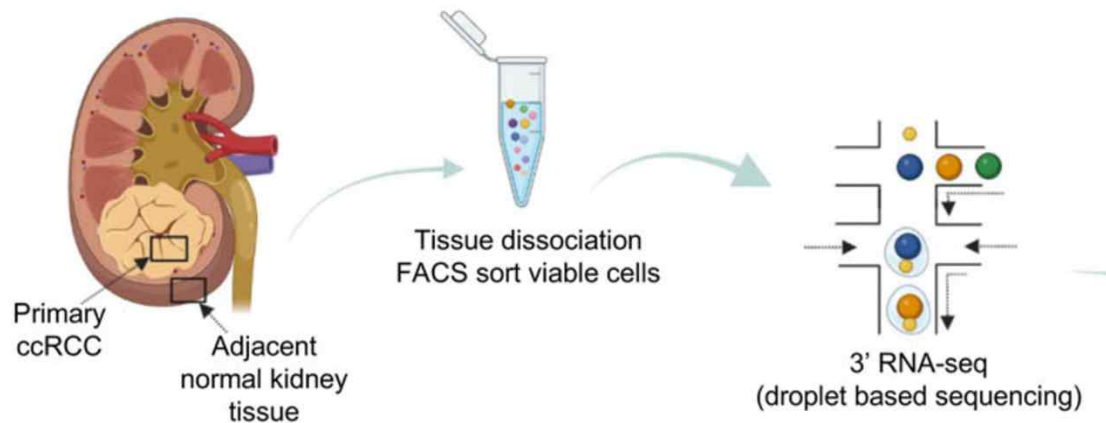


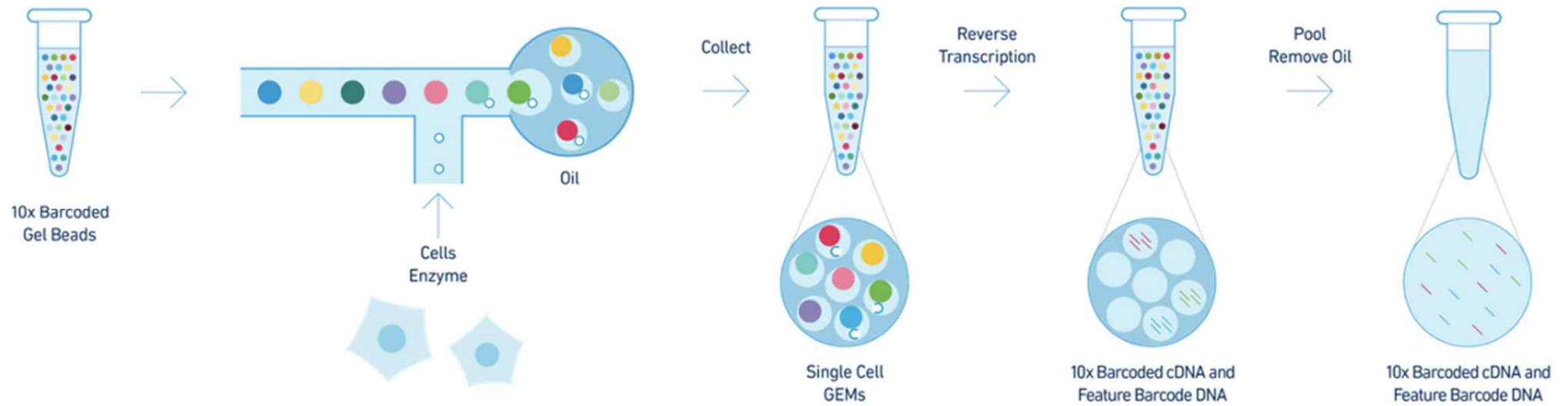
A transcriptional metastatic signature predicts survival in clear cell renal cell carcinoma



ccRCC 3, normal 3

PR1-Normal PR1-tumor PR2-Normal PR2-tumor PR6-Normal PR6-tumor

10x genomics



Goal: To learn how to pre-process and analyze single-cell RNA sequencing data with real data

Content

- 0) Setup and data downloading
- 1) Data importing
- 2) Quality control
- 3) Normalization & Scaling
- 4) Feature selection
- 5) Dimension reduction (PCA & UMAP)
- 6) Batch correction (Harmony)
- 7) Clustering
- 8) Celltype annotation
- 9) DEG between groups
- 10) Signature analysis

Let's start!

R installation

[R: The R Project for Statistical Computing](https://www.r-project.org/)



[\[Home\]](#)

Download

[CRAN](#)

R Project

[About R](#)
[Logo](#)
[Contributors](#)
[What's New?](#)
[Reporting Bugs](#)
[Conferences](#)
[Search](#)
[Get Involved: Mailing Lists](#)
[Get Involved: Contributing](#)
[Developer Pages](#)
[R Blog](#)

R Foundation

[Foundation](#)
[Board](#)
[Members](#)

The R Project for Statistical Computing

Getting Started

R is a free software environment for statistical computing and graphics. It compiles and runs on a wide variety of UNIX platforms, Windows and MacOS. To [download R](#), please choose your preferred [CRAN mirror](#).

If you have questions about R like how to download and install the software, or what the license terms are, please read our [answers to frequently asked questions](#) before you send an email.

News

- [R version 4.5.0 \(How About a Twenty-Six\) prerelease versions](#) will appear starting Tuesday 2025-03-11. Final release is scheduled for Friday 2025-04-11.
- [R version 4.4.3 \(Trophy Case\)](#) has been released on 2025-02-28.
- The [useR! 2025](#) conference will take place at Duke University, in Durham, NC, USA, August 8-10.
- We are deeply sorry to announce that our friend and colleague Friedrich (Fritz) Leisch has died. [Read our tribute to Fritz here.](#)
- [R version 4.3.3 \(Angel Food Cake\)](#) (wrap-up of 4.3.x) was released on 2024-02-29.
- You can support the R Foundation with a renewable subscription as a [supporting member](#).

News via Mastodon

<https://cran.stat.dmpu.ty>

Japan

<https://ftp.yz.yamagata-u.ac.jp/pub/cran/>

Korea

<https://cran.yu.ac.kr/>

Mexico

<https://cran.itam.mx/>

<https://www.est.colpos.mx/>

Morocco

<https://mirror.marwan.ma/cran/>

Download and Install R

Precompiled binary distributions of the base system and contributed packages, **Windows and Mac** users most likely want one of these versions of R:

- [Download R for Linux \(Debian, Fedora/Redhat, Ubuntu\)](#)
- [Download R for macOS](#)
- [Download R for Windows](#)

R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above

Window

R for Windows

Subdirectories:

[base](#)

Binaries for base distribution. This is what you want to [install R for the first time](#).

[contrib](#)

Binaries of contributed CRAN packages (for R $\geq$ 4.0.x).

[old contrib](#)

Binaries of contributed CRAN packages for outdated versions of R (for R $<$ 4.0.x).

[Rtools](#)

Tools to build R and R packages. This is what you want to build your own packages c
itself.

MAC ??

Latest release:

For Apple silicon (M1,2,...) Macs:

[R-4.4.3-arm64.pkg](#)

5a60f8b6c89073
(ca. 97MB, notarized and signed)

R 4.4.3 binary for macOS 11 (**Big Sur**) and higher, signed and notarized packages.

Contains R 4.4.3 framework, R.app GUI 1.81, Tcl/Tk 8.6.12 X11 libraries and Texinfo 6.8. The latter two components are optional and can be omitted when choosing "custom install", they are on

R-4.4.3 for Windows

[Download R-4.4.3 for Windows](#) (85 megabytes, 64 bit)

[README on the Windows binary distribution](#)

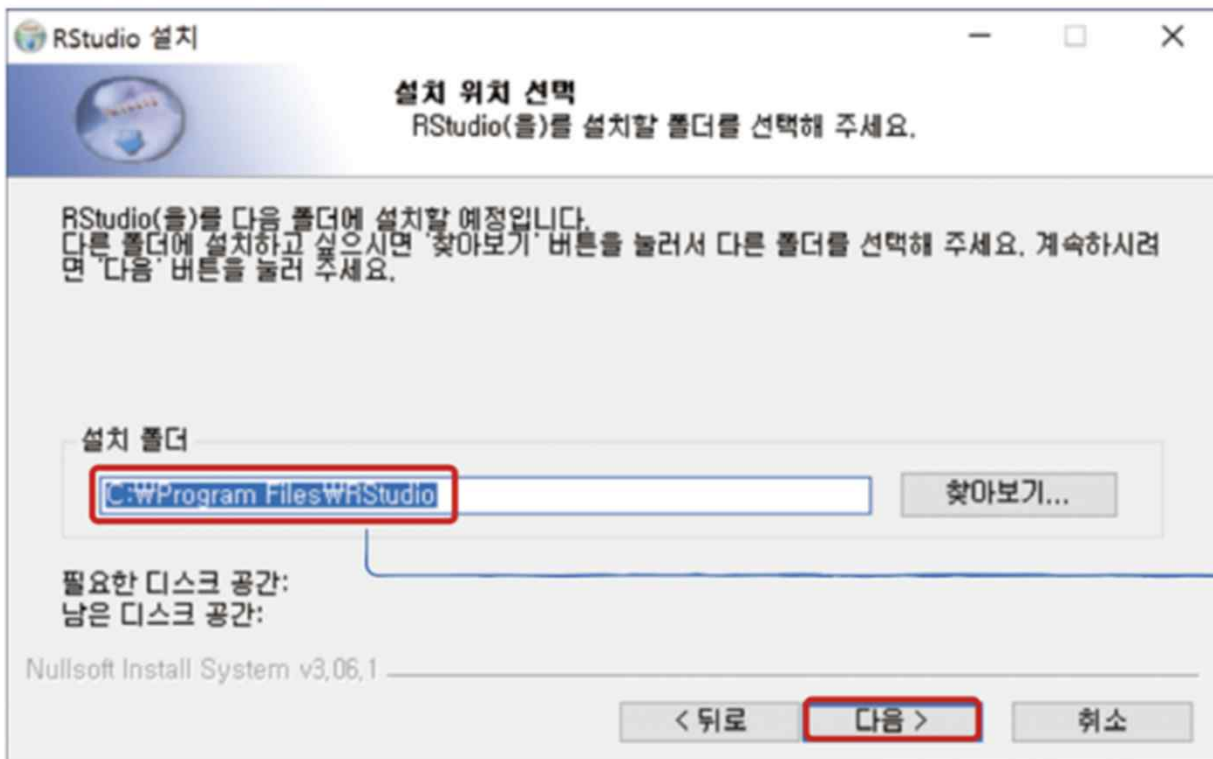
[New features in this version](#)

Download and install it
(default parameter setting)

Rstudio installation

<https://posit.co/download/rstudio-desktop/>

OS	Download	Size	SHA-256
Windows 10/11	RSTUDIO-2024.12.1-563.EXE 	265.28 MB	BB369743
macOS 13+	RSTUDIO-2024.12.1-563.DMG 	557.15 MB	BE73D3A9
Ubuntu 20/Debian 11	RSTUDIO-2024.12.1-563-AMD64.DEB 	203.14 MB	EE259A88



경로와 폴더 이름에는
한글이 없어야 합니다.

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function

Addins

Project: (None)

Console Terminal x Background Jobs x

R 4.4.3 · ~/

R version 4.4.3 (2025-02-28 ucrt) -- "Trophy Case"
Copyright (c) 2025 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

R은 자유 소프트웨어이며, 어떠한 형태의 보증없이 배포됩니다.
또한, 일정한 조건하에서 이것을 재배포 할 수 있습니다.
배포와 관련된 상세한 내용은 'license()' 또는 'licence()'을 통하여 확인할 수 있습니다.

R은 많은 기여자들이 참여하는 공동프로젝트입니다.
'contributors()'라고 입력하시면 이에 대한 더 많은 정보를 확인할 수 있습니다.
그리고, R 또는 R 패키지들을 출판물에 인용하는 방법에 대해서는 'citation()'을 통해 확인할 수 있습니다.

'demo()'를 입력하신다면 몇가지 데모를 보실 수 있으며, 'help()'를 입력하시면 온라인 도움말을 이용하실 수 있습니다.
또한, 'help.start()'의 입력을 통하여 HTML 브라우저에 의한 도움말을 이용하실 수 있습니다.
R의 종료를 원하시면 'q()'를 입력해주세요.

> |

Environment History Connections Tutorial

Import Dataset 92 MiB

Global Environment

Environment is empty

Files Plots Packages Help Viewer Presentation

Folder File Delete Rename

Home

	Name	Size	Modified
☐	.Rhistory	20 B	Mar 13, 2025, 11:21 AM
☐	'fmLj7g.jpg	228 KB	Mar 13, 2025, 8:47 AM
☐	사용자 지정 Office 서식 파일		
☐	카카오톡 받은 파일		
☐	Default.rdp	0 B	Feb 28, 2025, 3:32 PM
☐	desktop.ini	402 B	Mar 4, 2025, 8:34 AM
☐	My Music		
☐	My Pictures		
☐	My Videos		
☐	Welcome to Hwp.hwp	26.5 KB	Jun 13, 2023, 7:56 AM
☐	Welcome to Word.hwd	14.8 KB	Jun 13, 2023, 7:56 AM

-Downloading R packages

R

```
install.packages('Seurat')
```

```
# https://satijalab.org/seurat/
```

```
install.packages("harmony")
```

```
# https://github.com/immunogenomics/harmony
```

```
install.packages("Matrix")
```

```
library(Seurat)
```

```
library(harmony)
```



Harmony

build error build failing DOI [10.1038/s41592-019-0619-0](https://doi.org/10.1038/s41592-019-0619-0)

Fast, sensitive and accurate integration of single-cell data with Harmony

Check out the manuscript in Nature Methods:

- [nature website](#)
- [read link](#)



-Data download

GEO: Gene Expression Omnibus

Gene Expression Omnibus

GEO is a public functional genomics data repository supporting MIAME-compliant data submissions. Array- and sequence-based data are accepted. Tools are provided to help users query and download experiments and curated gene expression profiles.



Keyword or GEO Accession

Getting Started

- Overview
- FAQ
- About GEO DataSets
- About GEO Profiles
- About GEO2R Analysis
- How to Construct a Query
- How to Download Data

Tools

- Search for Studies at GEO DataSets
- Search for Gene Expression at GEO Profiles
- Search GEO Documentation
- Analyze a Study with GEO2R
- Studies with Genome Data Viewer Tracks
- Programmatic Access
- FTP Site

Browse Content

Repository Browser

DataSets:	4348
Series: 	248956
Platforms:	27129
Samples:	7708425

Samples (4) [Less...](#)

- [GSM4598898](#) tumor-lung_gex
- [GSM4598899](#) normal-lung_gex
- [GSM4598900](#) tumor-lung_vdj
- [GSM4598901](#) normal-lung_vdj

Supplementary file	Size	Download	File type/resource
GSM4598898_tumor_gex_barcode.svs.gz	37.7 Kb	(ftp) (http)	TSV
GSM4598898_tumor_gex_features.svs.gz	272.8 Kb	(ftp) (http)	TSV
GSM4598898_tumor_gex_matrix.mtx.gz	45.9 Mb	(ftp) (http)	MTX

Change names into

barcode.svs.gz

features.svs.gz

matrix.mtx.gz

→ own folder for each sample

```
PR1-Normal/:
barcode.svs.gz  features.svs.gz  matrix.mtx.gz

PR1-tumor/:
barcode.svs.gz  features.svs.gz  matrix.mtx.gz

PR2-Normal/:
barcode.svs.gz  features.svs.gz  matrix.mtx.gz
```

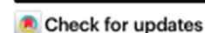


A transcriptional metastatic signature predicts survival in clear cell renal cell carcinoma

Received: 11 November 2021

Accepted: 13 September 2022

Published online: 30 September 2022



Adele M. Alchahin^{1,11}, Shenglin Mei^{2,11}✉, Ioanna Tsea¹, Taghreed Hirz^{3,4,5},
Yumna Kfoury^{3,4,5}, Douglas Dahl⁶, Chin-Lee Wu⁷, Alexander O. Subtelny⁷,
Shulin Wu⁷, David T. Scadden^{3,4,5}, John H. Shin⁸, Philip J. Saylor^{9,11},
David B. Sykes^{3,4,5,11}, Peter V. Kharchenko^{2,4,10,11}✉ &
Ninib Baryawno^{1,3,4,5,11}✉

Data availability

Raw single-cell RNA-sequencing data and processed data are publicly available and can be accessed from the NCBI Gene Expression Omnibus database under accession code **GSE178481**. GRCh38 human reference genome was download from 10X genomics [<https://support.10xgenomics.com/single-cell-gene-expression/software/downloads/>]. For the joint alignment analysis with public ccRCC scRNA-seq data, we downloaded raw count matrix and cell annotation from European Genome-phenome Archive under controlled access: [EGAS00001002171](https://www.ebi.ac.uk/ena/browser/view/EGAS00001002171), [EGAS00001002486](https://www.ebi.ac.uk/ena/browser/view/EGAS00001002486), [EGAS00001002325](https://www.ebi.ac.uk/ena/browser/view/EGAS00001002325), and Single Cell Portal [https://singlecell.broadinstitute.org/single_cell/study/SCP1288/]^{11,13}. 10X Visium Spatial Transcriptomics (ST) data were download from [GSE175540](https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE175540)²⁵. For bulk RNA-seq data, TCGA clear cell renal cell carcinoma (KIRC) cohort were downloaded from the cBioPortal [https://www.cbioportal.org/study/clinicalData?id=kirc_tcga]. The Checkmate 025 cohort data were obtained from the supplementary table of published results¹¹ (<https://doi.org/10.1038/s41591-020-0839-y>). Source data are provided with this paper.

HOMESEARCHSITE MAP

GEO PublicationsFAQMIAMEEmail GEO

NCBI > GEO > Accession Display ?

Not logged in | Login ?

Scope: SelfFormat: HTMLAmount: QuickGEO accession: GSE178481GO

Series GSE178481

Query DataSets for GSE178481

Status

Public on Aug 18, 2022

Title

A transcriptional metastatic signature predicts survival in clear cell renal cell carcinoma

Organism

Homo sapiens

Experiment type

Expression profiling by high throughput sequencing

Summary

Novel single cell technologies have paved the way for the characterization and improved biological understanding of clear cell renal cell carcinoma (ccRCC) where the focus has been mostly on the immunological compartments of the tumor microenvironment (TME). However, the identification of metastatic tumor cell clones, and the stromal compartments of TME of untreated ccRCC patients still remains to be elucidated. Here we perform single-cell RNA-sequencing on treatment naive ccRCC and adjacent normal kidney tissue.

Overall design

Human kidney tissue were disaggregated and profiled using the Chromium Single Cell 3' Library and Gel Bead Kit v2 (10X Genomics) .

Web link

https://pubmed.ncbi.nlm.nih.gov/36180422/

Contributor(s)

Mei S

Citation(s)

Alchahin AM, Mei S, Tsea I, Hirz T et al. A transcriptional metastatic signature predicts survival in clear cell renal cell carcinoma. *Nat Commun* 2022 Sep 30;13(1):5747. PMID: 36180422
Mei S, Alchahin AM, Tsea I, Kfoury Y et al. Single-cell analysis of immune and stroma cell remodeling in clear cell renal cell carcinoma primary tumors and bone metastatic lesions. *Genome Med* 2024 Jan 29;16(1):1. PMID: 38281962

- Platforms (1)

GPL18573 Illumina NextSeq 500 (Homo sapiens)
- Samples (26)

Less...

GSM5392398

RCC-PR6-Normal

GSM5392399

RCC-PR6-PTumor

GSM5392400

RCC-PR8-Normal

GSM5392401

RCC-PR7-PTumor

GSM5392402

RCC-PR9-Normal

GSM5392403

RCC-PR9-PTumor

GSM5392404

RCC-PR9-PTumor2

GSM5392405

RCC-PR5-Normal

GSM5392406

RCC-PR5-PTumor1

GSM5392407

RCC-PR5-PTumor2

Supplementary file	Size	Download	File type/resource
GSM5392398_RCC-PR6-Normal.count.csv.gz	2.6 Mb	(ftp)(http)	CSV

SRA Run Selector ?

RCC-PR1-Normal
RCC-PR1-PTumor

RCC-PR2-Normal
RCC-PR2-PTumor

RCC-PR6-Normal
RCC-PR6-PTumor

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function

Console Terminal Background Jobs

R 4.4.3 ~ /

R version 4.4.3 (2025-02-28 ucrt) -- "Trophy Case"

Copyright (C) 2025 The R Foundation for Statistical Computing

Platform: x86_64-w64-mingw32/x64

R은 자유 소프트웨어이며, 어떠한 형태의 보증없이 배포됩니다.

또한, 일정한 조건하에서 이것을 재배포 할 수 있습니다.

배포와 관련된 상세한 내용은 'license()' 또는 'licence()'을 통하여 확인할 수 있습니다.

R은 많은 기여자들이 참여하는 공동프로젝트입니다.

'contributors()'라고 입력하시면 이에 대한 더 많은 정보를 확인하실 수 있습니다.

그리고, R 또는 R 패키지들을 출판물에 인용하는 방법에 대해서는 'citation()'을 통해 확인하시길 부탁드립니다.

'demo()'를 입력하신다면 몇가지 데모를 보실 수 있으며, 'help()'를 입력하시면 온라인 도움말을 이용하실 수 있습니다.

또한, 'help.start()'의 입력을 통하여 HTML 브라우저에 의한 도움말을 이용하실 수 있습니다.

R의 종료를 원하시면 'q()'를 입력해주세요.

> |

Put your files into here →

문서

백업 시작 > ... 사용자 > admin > 문서

새로 만들기 > Import Data

Environment History

R Global Environment

다운로드 문서 사진 음악 동영상 perturbation 논문

이름

수정한 날짜

유형

사용자 지정 Office 서식 파일

카카오톡 받은 파일

.Rhistory

'fmLj7g'

Welcome to Hwp

Welcome to Word

GSM5392398_RCC-PR6-Normal.count.c...

2025-02-19 오전 11:26

2025-03-05 오후 2:13

2025-03-13 오전 11:33

2025-03-13 오후 1:06

2023-06-13 오전 7:56

2023-06-13 오전 7:56

2025-03-13 오후 1:17

파일 폴더

파일 폴더

RHISTORY 파일

JPG 파일

한컴오피스 한글 ...

한컴오피스 한워...

압축된 보관 폴더

Files Plots Packages

Folder File

Home

Name

.Rhistory

'fmLj7g.jpg'

사용자 지정 Office 서식 파일

카카오톡 받은 파일

Default.rdp

desktop.ini

My Music

My Pictures

My Videos

Welcome to Hwp.hwp

Welcome to Word.hwdt

215 B

213.4 KB

0 B

402 B

26.5 KB

14.8 KB

Mar 13, 2025, 11:33 AM

Mar 13, 2025, 1:06 PM

Feb 28, 2025, 3:32 PM

Mar 4, 2025, 8:34 AM

Jun 13, 2023, 7:56 AM

Jun 13, 2023, 7:56 AM

```
ls *.gz > lecture_list
```

해당 명령어가 안될 경우

Terminal에서
텍스트 파일 만들어서 파일 이름 작성하기

```
GSM5392398_RCC-PR6-Normal.count.csv.gz  
GSM5392399_RCC-PR6-PTumor.count.csv.gz  
GSM5392414_RCC-PR1-Normal.count.csv.gz  
GSM5392415_RCC-PR1-PTumor.count.csv.gz  
GSM5392416_RCC-PR2-Normal.count.csv.gz  
GSM5392417_RCC-PR2-PTumor.count.csv.gz
```



```
#install.packages("Matrix")
library(Matrix)
data_list <- readLines("lecture_list")

for (j in c(1:length(data_list))) {
  data <- data_list[j]
  print(data)
  sample <- strsplit(data, "RCC-")
  sample1 <- strsplit(sample[[1]][2], ".count")[[1]][1]
  group <- strsplit(sample1, "-")[[1]][2]

  a <- read.csv(data, check.names = F)
  a1 <- a[, 2:dim(a)[2]]
  rownames(a1) <- a[, 1]

  cells <- colnames(a1)
  g1 <- rownames(a1)
  mat <- as(as.matrix(a1), "sparseMatrix")
  genes <- rownames(a1)
  genes <- cbind(genes, genes)

  system(paste("mkdir ", sample1, sep = ""))
  setwd(sample1)
  write.table(cells, "barcodes.tsv", quote = FALSE, row.names = FALSE, col.names = FALSE)
  write.table(genes, "features.tsv", quote = FALSE, row.names = FALSE, col.names = FALSE, sep = "\t")
  writeMM(mat, "matrix.mtx")
  setwd("../")
}
```

텍스트 파일을 만들었을 경우
lecture_list.txt



```
system(paste("mkdir ", sample1, sep = ""))
```

명령어가 안될 경우

setwd(sample1)에서 error가 발생할 것임

Terminal에서

mkdir PR6-Normal

mkdir PR6-PTumor

mkdir PR1-Normal

mkdir PR1-PTumor

mkdir PR2-Normal

mkdir PR2-PTumor

```
#install.packages("Matrix")
library(Matrix)
data_list <- readLines("lecture_list")

for (j in c(1:length(data_list))){
  data <- data_list[j]
  print(data)
  sample <- strsplit(data, "RCC-")
  sample1<-strsplit(sample[[1]][2], ".count")[[1]][1]
  group <- strsplit(sample1, "-")[[1]][2]

  a<-read.csv(data, check.names = F)
  a1<-a[,2:dim(a)[2]]
  rownames(a1)<-a[,1]

  cells <- colnames(a1)
  g1 <- rownames(a1)
  mat <- as(as.matrix(a1), "sparseMatrix")
  genes <- rownames(a1)
  genes <- cbind(genes, genes)

  system(paste("mkdir ", sample1, sep = ""))
  setwd(sample1)
  write.table(cells, "barcodes.tsv", quote = FALSE, row.names = FALSE, col.names = FALSE)
  write.table(genes, "features.tsv", quote = FALSE, row.names = FALSE, col.names = FALSE, sep = "\t")
  writeMM(mat, "matrix.mtx")
  setwd("../")
}
```

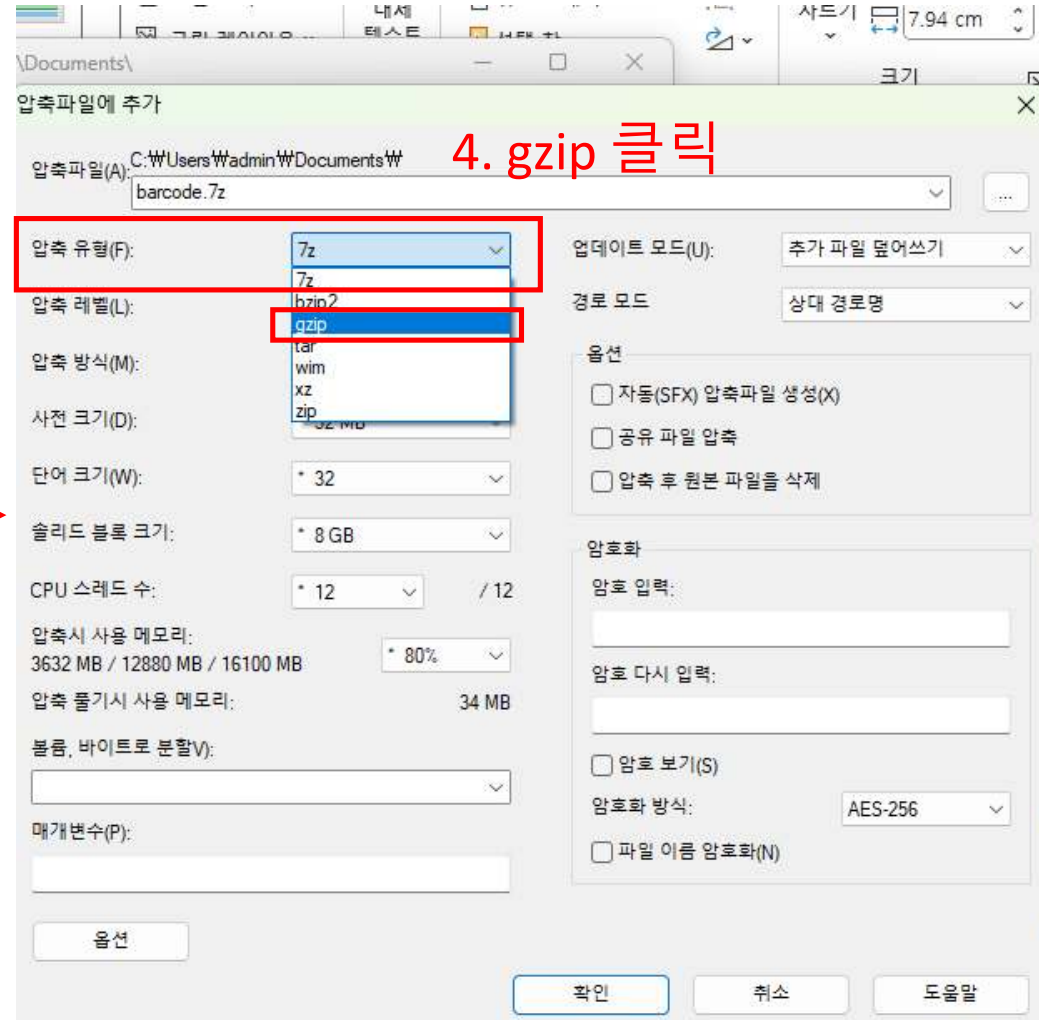
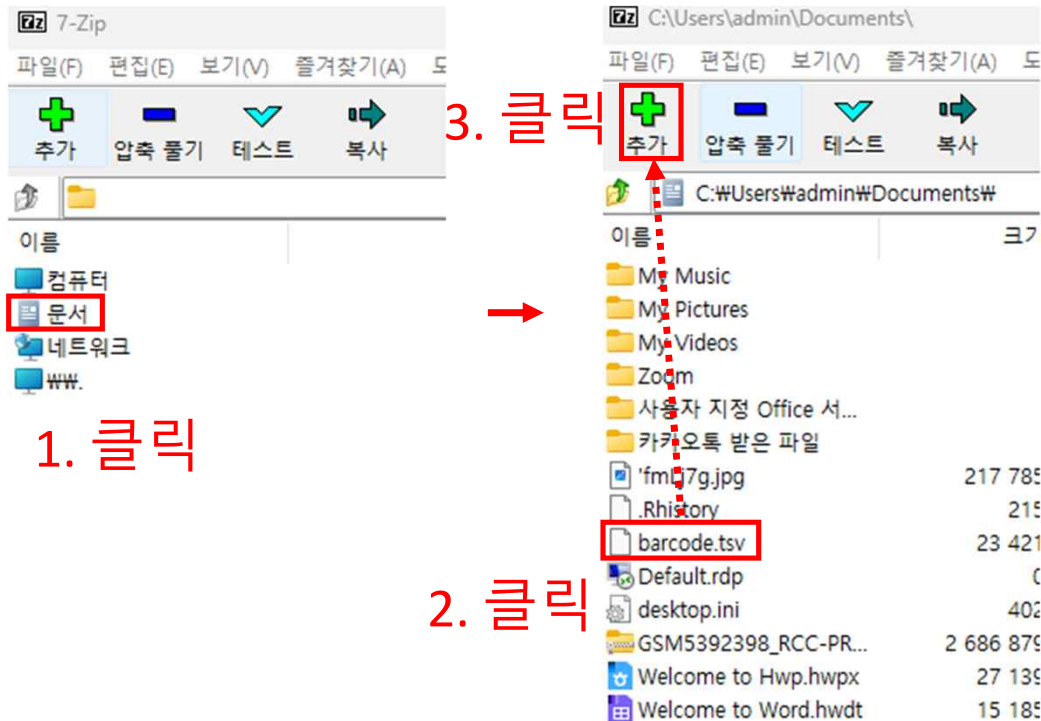
이 부분 삭제

Terminal에서
gzip [파일명]

gzip barcodes.tsv
gzip features.tsv
gzip matrix.mtx

barcodes.tsv.gz
features.tsv.gz
matrix.mtx.gz

Terminal에서 gzip이 안될 경우
구글에서 7-zip을 다운 받기
앱 실행
해당 파일 찾아서 gz 압축하기



일반적으로 GEO에 데이터를 받을 때 3개의 파일 형태로 주어짐 →
꼭! 파일명을 아래와 같이 바꿔주어야 함!
이외의 형식으로 파일을 제공해줄 경우 본인의 능력에 맞춰서
데이터를 확보하기 바람.

Supplementary file	Size	Download	File type/resource
GSM4598898_tumor_gex_barcodes.tsv.gz	37.7 Kb	(ftp) (http)	TSV
GSM4598898_tumor_gex_features.tsv.gz	272.8 Kb	(ftp) (http)	TSV
GSM4598898_tumor_gex_matrix.mtx.gz	45.9 Mb	(ftp) (http)	MTX

Change names into
barcodes.tsv.gz
features.tsv.gz
matrix.mtx.gz
→ own folder for each sample

1) Data importing

Why? We should make the object into analyzable format

min.cells: how many cells are expressing that gene

min.features: how many genes are expressed in single-cell (too low → it won't be a cell; empty droplet)

```
PR1-Normal/:
barcodes.tsv.gz  features.tsv.gz  matrix.mtx.gz
PR2-tumor/:
barcodes.tsv.gz  features.tsv.gz  matrix.mtx.gz
PR1-tumor/:
barcodes.tsv.gz  features.tsv.gz  matrix.mtx.gz
PR6-Normal/:
barcodes.tsv.gz  features.tsv.gz  matrix.mtx.gz
PR2-Normal/:
barcodes.tsv.gz  features.tsv.gz  matrix.mtx.gz
PR6-tumor/:
barcodes.tsv.gz  features.tsv.gz  matrix.mtx.gz
```

Script

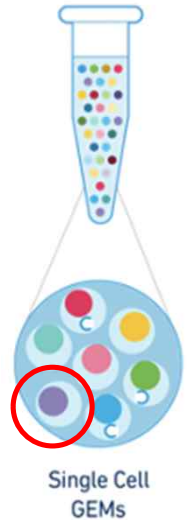
```
data <- Read10X(data.dir = "PR1-Normal/") #PR1-Normal
```

```
ccrcc <- CreateSeuratObject(counts = data, min.cells = 0, min.features = 0)
```

```
ccrcc$sample <- "PR1-Normal"
```

```
ccrcc$group <- "normal"
```

Metadata



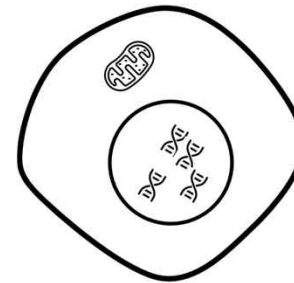
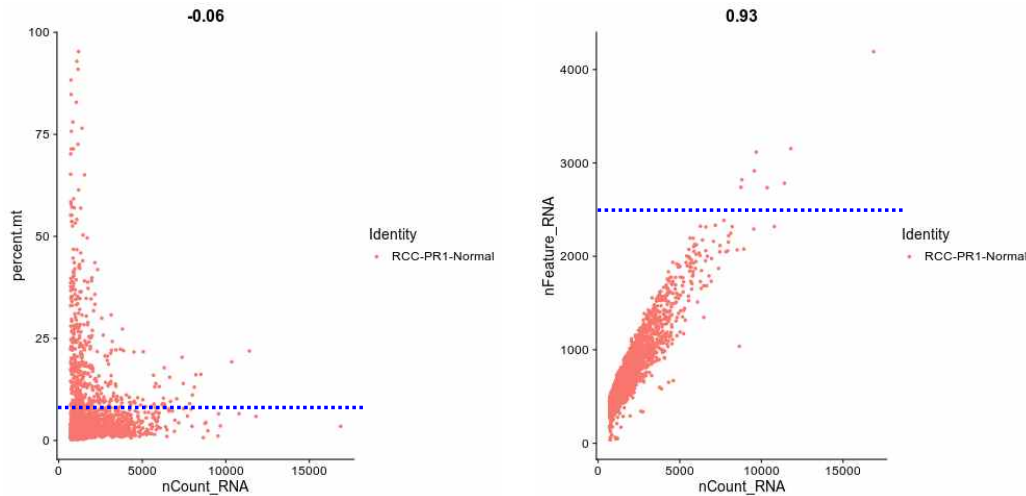
2) Quality control

Why? There might be some bad cells collected
(reason: cell stress during processing, high drop-out)

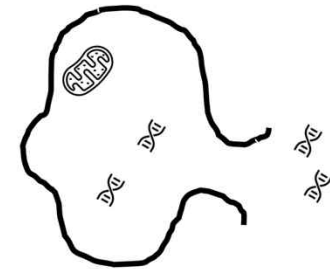
Low nCount or nFeature → empty droplet

High nCount or nFeature → doublet

High MT % → dead cell



Good!



Dead cell

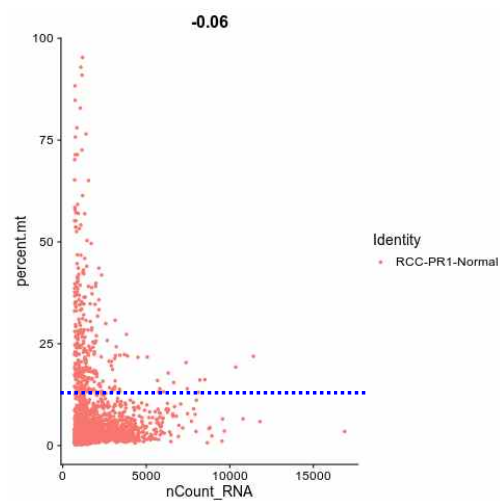
Script

```
ccrcc[["percent.mt"]] <- PercentageFeatureSet(ccrcc, pattern = "^MT-")  
FeatureScatter(ccrcc, feature1 = "nCount_RNA", feature2 = "percent.mt")
```

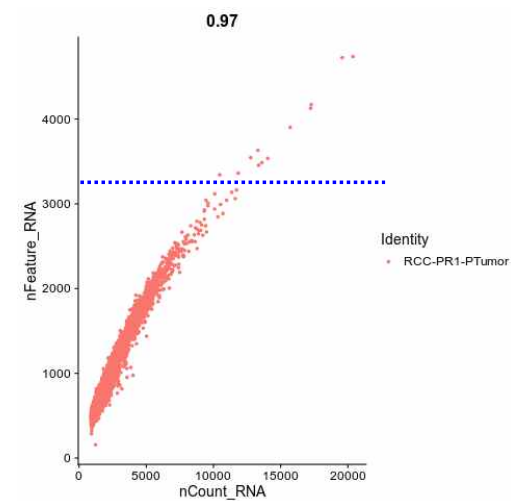
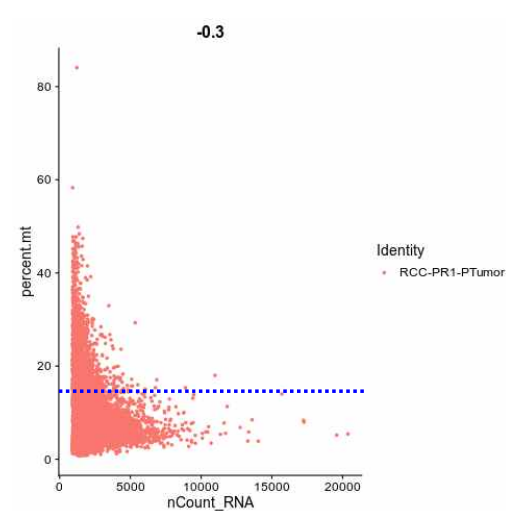
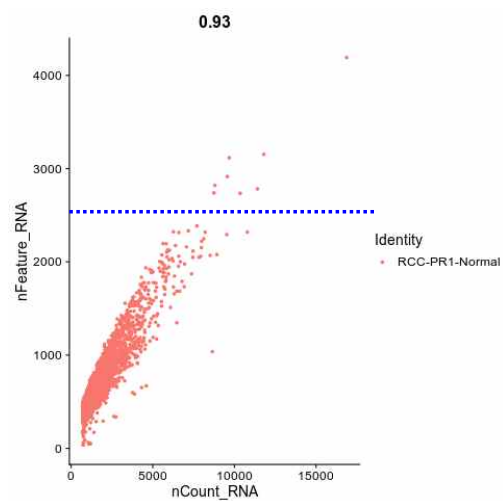
```
FeatureScatter(ccrcc, feature1 = "nCount_RNA", feature2 = "nFeature_RNA")
```

```
ccrcc <- subset(ccrcc, subset = percent.mt < 12.5 & nFeature_RNA < 2500 & nFeature_RNA > 200)
```

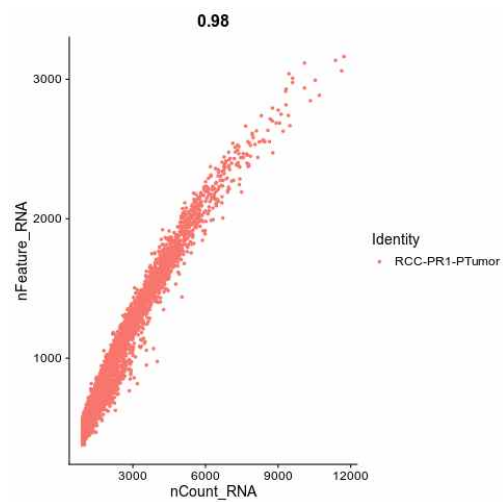
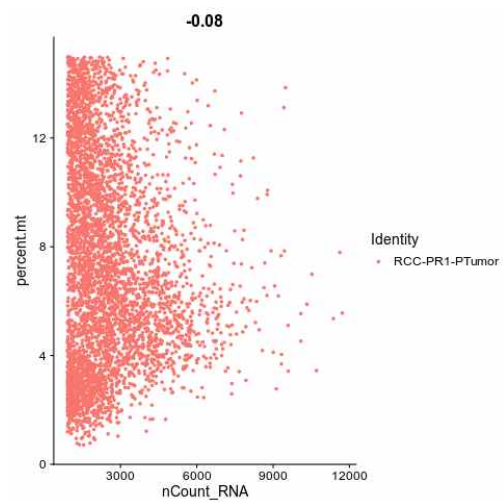

PR1-Normal (12.5, 2500)



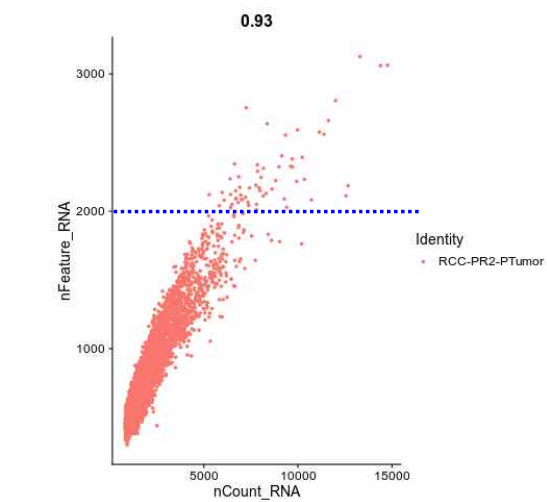
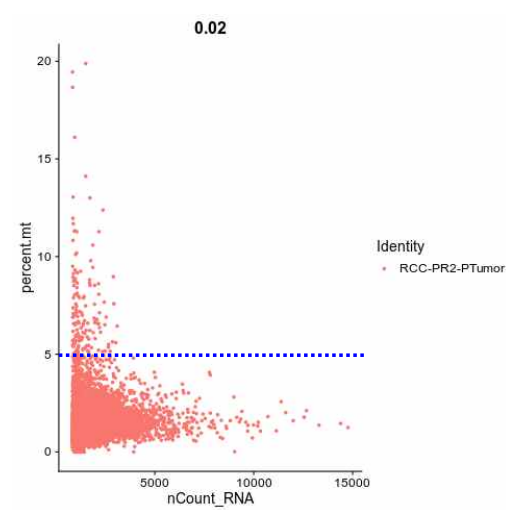
PR1-tumor (15, 3200)



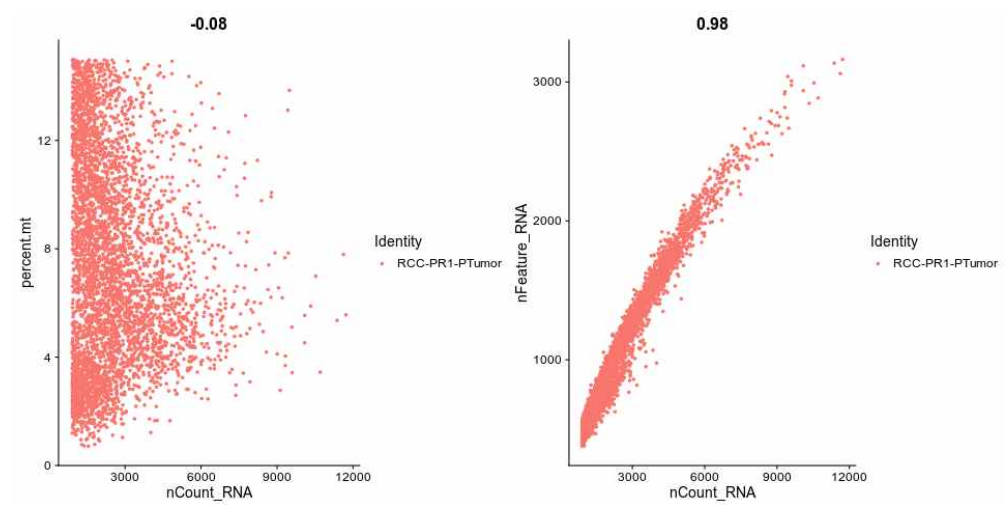
PR2-Normal (-)



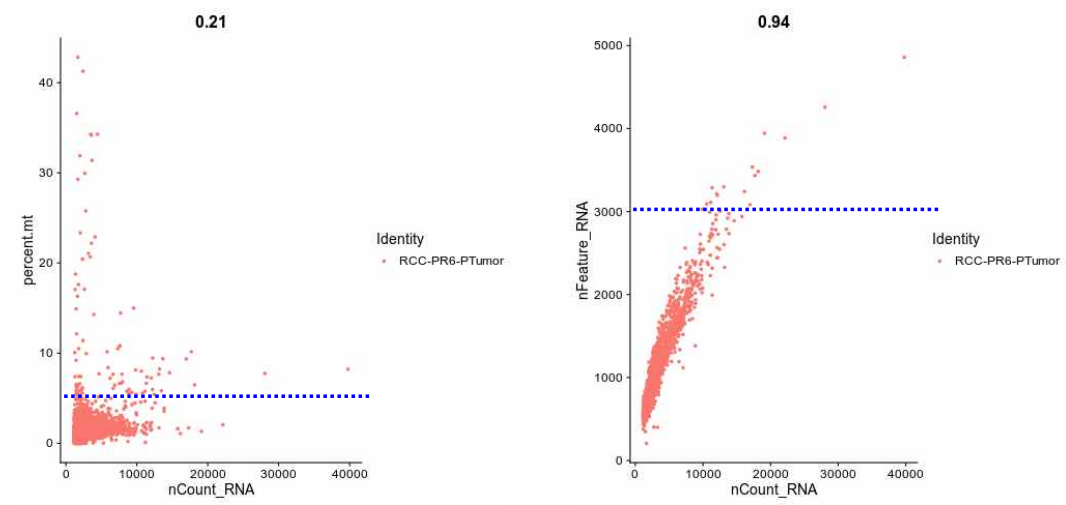
PR2-tumor (5, 2000)



PR6-Normal (-)



PR6-tumor (5, 3000)



3) Normalization & Scaling

Why?

Total read count normalization: Adjust read-depth between each cell

Log-transformation: adjust the variance of the gene expression matrix

Scaling: gives equal weight in downstream analyses, so that highly-expressed genes do not dominate

Script

```
ccrcc <- NormalizeData(ccrcc, normalization.method = "LogNormalize", scale.factor = 10000)
```

```
ccrcc <- ScaleData(ccrcc)
```

4) Feature selection

Why? scRNA-seq tends to have a lot of drop-out

→ Not all the gene is informative → select informative genes and reduce noise

```
> VariableFeatures(ccrcc1)[1:20]
[1] "MT1G"      "DEFB1"      "S100A2"      "ALDOB"      "S100A9"      "S100A8"
[7] "MT1H"      "FXYD4"      "MTRNR2L12"   "MIOX"       "MTRNR2L8"    "UMOD"
[13] "WFDC2"     "IGLL5"      "MZB1"        "GPX3"       "FXYD2"       "CCL3"
[19] "TAGLN"     "CXCL14"
```

Script

```
ccrcc <- FindVariableFeatures(ccrcc, selection.method = "vst", nfeatures = 2000)
```

```
variable_list <- VariableFeatures(ccrcc)
```

5) Dimension reduction (PCA & UMAP)

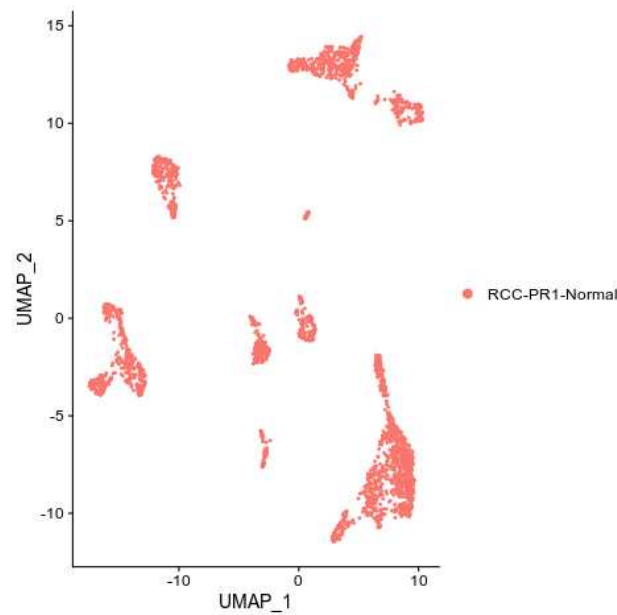
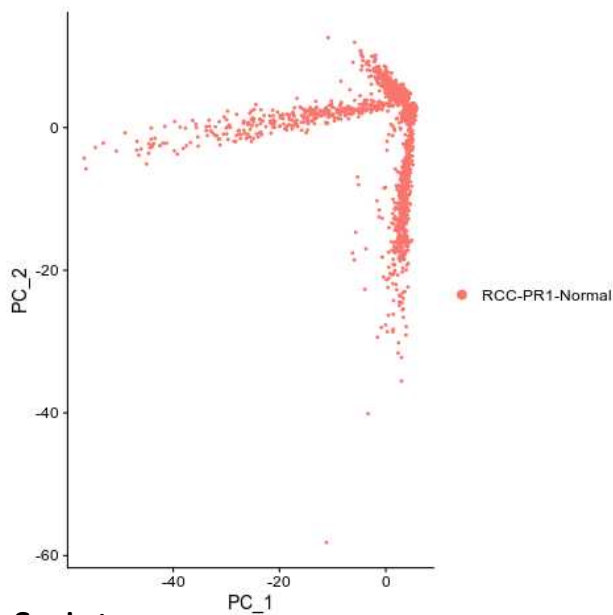
Why? Too many features (genes) → hard to interpret

→ Dimension reduction: abstract of many features!

PCA: commonly used in bulk RNA-seq data → insufficient for scRNA-seq

UMAP: adjusted for scRNA-seq (similar cells to be close and different cells to be far away)

dims: how many axis of PCs to use



Script

```
ccrcc <- RunPCA(ccrcc, npcs = 30)
```

```
ccrcc <- RunUMAP(ccrcc, dims = 1:30)
```

```
DimPlot(ccrcc)
```

6) Batch correction (Harmony)

Script

```
total<-merge(ccrcc1,y= list(ccrcc2,ccrcc3,ccrcc4,ccrcc5,ccrcc6))
```

```
total <- JoinLayers(total)
```

```
total<-NormalizeData(total)
```

```
total<- ScaleData(total)
```

```
total<- FindVariableFeatures(total, selection.method = "vst", nfeatures = 2000)
```

```
total<- RunPCA(total, npcs = 30)
```

```
total <- RunUMAP(total, dims = 1:30)
```

```
DimPlot(total, group.by = "sample")
```

```
library(harmony)
```

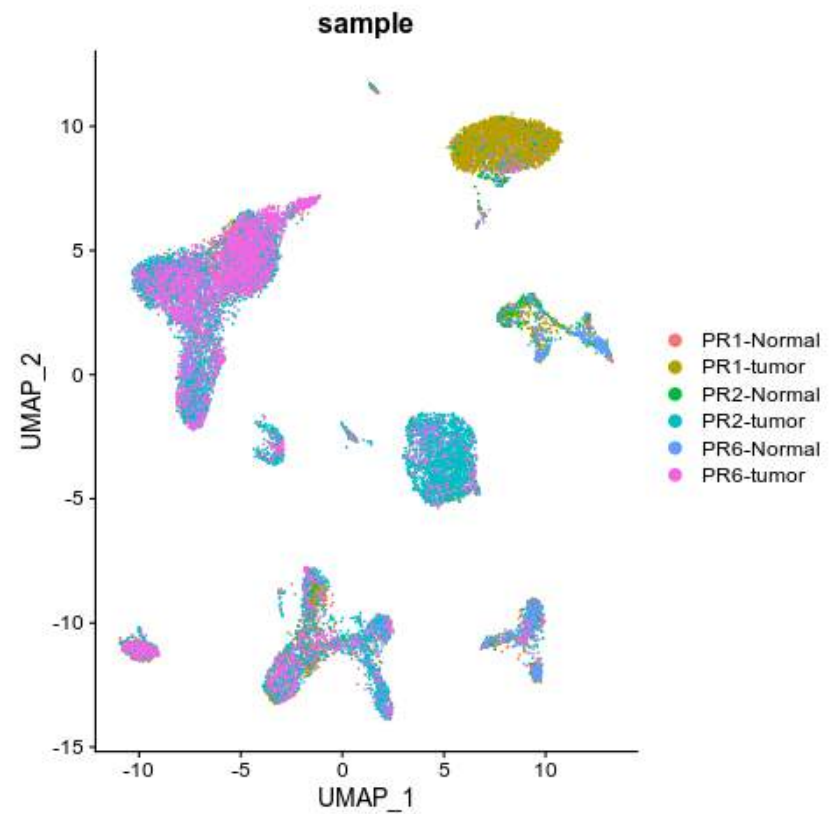
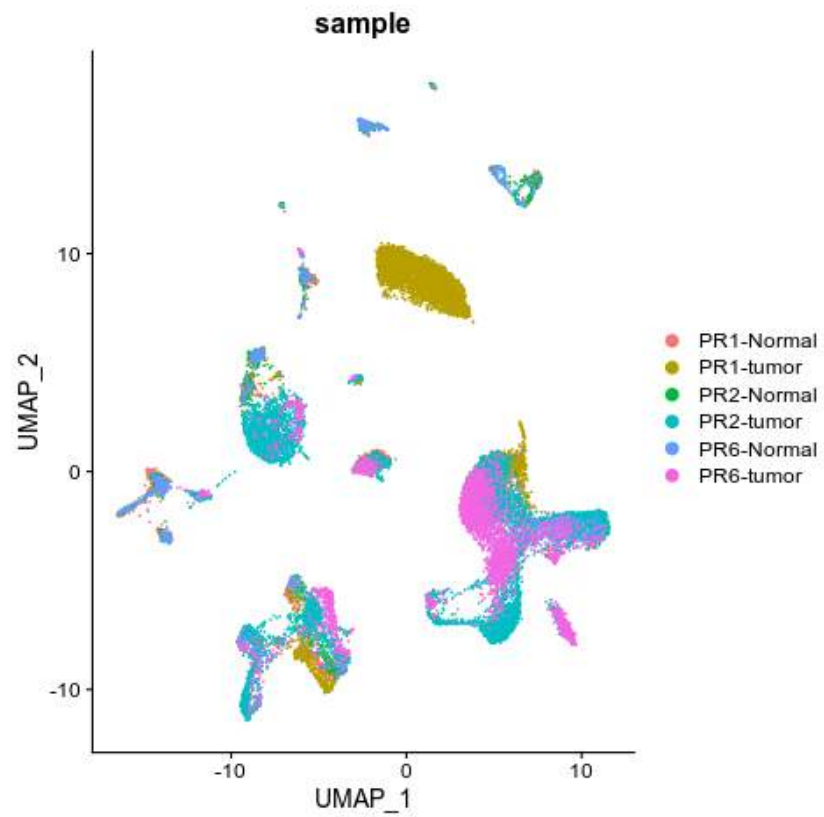
```
total <-RunHarmony(total, c("sample"), dims.use = 1:30, plot_convergence=F)
```

```
total <- RunUMAP(total, reduction= "harmony", dims = 1:30)
```

```
DimPlot(total, group.by = "sample")
```

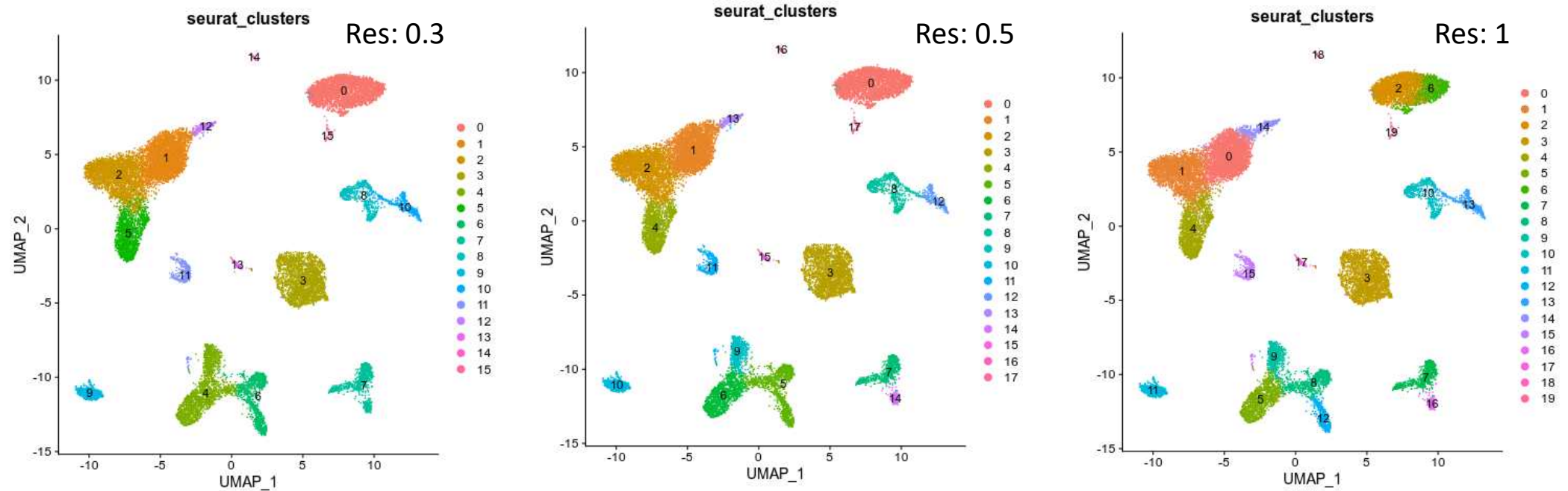
6) Batch correction (Harmony)

Why? To remove technical variation between samples



7) Clustering

Resolution → controls the number of clusters



Script

```
total <- FindNeighbors(total, reduction = "harmony", dims = 1:30)
total <- FindClusters(total, resolution = 0.3)
DimPlot(total, group.by = "seurat_clusters", label = T)
```

8) Celltype annotation

How? Distribution of marker gene expression

* Immune cell

Tcell: "CD3D", "CD3E"

CD4 T cell: "CD4"

CD8 T cell: "CD8A", "CD8B"

Treg: "FOXP3", "IL2RA"

NK cell: "KLRB1", "GNLY", "KLRD1", "NKG7"

B cell: "MS4A1", "CD79B"

Macrophage: "C1QA", "C1QB", "CD14", "CD68"

Monocyte: "FCN1", "S100A8", "S100A9"

Mast: "TPSAB1", "CPA3"

Cycling: "MKI67", "TOP2A"

* Non-immune cell

Pericyte: "CSPG4", "MCAM", "MYH11"

Endothelial cell: "RAMP2", "RNASE1", "ENG", "EGFL7"

Cancer: "PAX8"

Epithelial cell: "SLC26A7", "EPCAM", "MUC1"

	p_val	avg_log2FC	pct.1	pct.2	p_val_adj	cluster	gene
CRYAB	0	4.405264	0.996	0.158	0	0	CRYAB
RBP4	0	3.934706	0.880	0.032	0	0	RBP4
UGT2B7	0	3.844966	0.950	0.039	0	0	UGT2B7
SPP1	0	3.729567	0.924	0.081	0	0	SPP1
WFDC2	0	3.711430	0.954	0.081	0	0	WFDC2
CLU	0	3.401655	0.915	0.089	0	0	CLU
DEFB1	0	3.202296	0.934	0.110	0	0	DEFB1
TNFRSF12A	0	2.974357	0.872	0.094	0	0	TNFRSF12A
PDZK1IP1	0	2.943278	0.761	0.028	0	0	PDZK1IP1
KRT18	0	2.863520	0.847	0.068	0	0	KRT18
TFPI2	0	2.857835	0.692	0.022	0	0	TFPI2
AC073218.2	0	2.825047	0.808	0.029	0	0	AC073218.2
SOSTDC1	0	2.822084	0.679	0.018	0	0	SOSTDC1
IGFBP6	0	2.698095	0.809	0.064	0	0	IGFBP6
CXCL14	0	2.682915	0.777	0.053	0	0	CXCL14
TMEM176A	0	2.603952	0.756	0.043	0	0	TMEM176A
SLPI	0	2.561452	0.519	0.020	0	0	SLPI
TSPAN1	0	2.555455	0.718	0.028	0	0	TSPAN1

8) Celltype annotation

Script

```
marker <- FindAllMarkers(total, only.pos = TRUE, min.pct = 0.25, logfc.threshold = 0.25)
```

marker table 보는 방법

head(marker)

#상단 5줄

tail(marker)

#하단 5줄

원하는 cluster만 보기

marker[marker\$cluster == "0",]

marker[marker\$cluster == "2",]

몇줄만 보기

marker[,c(1:10)]

marker[,c(3:20)]

0 클러스터 보기

2 클러스터 보기

1줄 부터 10줄

3줄 부터 20줄

	p_val	avg_log2FC	pct.1	pct.2	p_val_adj	cluster	gene
CRYAB	0	4.405264	0.996	0.158	0	0	CRYAB
RBP4	0	3.934706	0.880	0.032	0	0	RBP4
UGT2B7	0	3.844966	0.950	0.039	0	0	UGT2B7
SPP1	0	3.729567	0.924	0.081	0	0	SPP1
WFDC2	0	3.711430	0.954	0.081	0	0	WFDC2
CLU	0	3.401655	0.915	0.089	0	0	CLU
DEFB1	0	3.202296	0.934	0.110	0	0	DEFB1
TNFRSF12A	0	2.974357	0.872	0.094	0	0	TNFRSF12A
PDZK1IP1	0	2.943278	0.761	0.028	0	0	PDZK1IP1
KRT18	0	2.863520	0.847	0.068	0	0	KRT18
TFPI2	0	2.857835	0.692	0.022	0	0	TFPI2
AC073218.2	0	2.825047	0.808	0.029	0	0	AC073218.2
SOSTDC1	0	2.822084	0.679	0.018	0	0	SOSTDC1
IGFBP6	0	2.698095	0.809	0.064	0	0	IGFBP6
CXCL14	0	2.682915	0.777	0.053	0	0	CXCL14
TMEM176A	0	2.603952	0.756	0.043	0	0	TMEM176A
SLPI	0	2.561452	0.519	0.020	0	0	SLPI
TSPAN1	0	2.555455	0.718	0.028	0	0	TSPAN1

8) Celltype annotation

Script

```
marker <- FindAllMarkers(total, only.pos = TRUE, min.pct = 0.25, logfc.threshold = 0.25)
```

```
#tcell NK
```

```
png("tcell.png", width = 1600, height = 1200)
```

```
FeaturePlot(total, features = c("CD3D", "CD3E", "CD4", "CD8A", "CD8B", "FOXP3",  
"IL2RA", "KLRB1", "GNLY", "KLRD1", "NKG7"))
```

```
dev.off()
```

```
#bcell, macrophage, mast, #DC, monocyte
```

```
png("bcell_myeloid.png", width = 1600, height = 1200)
```

```
FeaturePlot(total, features = c("MS4A1", "CD79B", "C1QA", "C1QB", "CD14", "CD68",  
"TPSAB1", "CPA3", "FCN1", "S100A8", "S100A9"))
```

```
dev.off()
```

8) Celltype annotation

```
#pericyte, endothelial cell
```

```
png("stroma.png", width = 1600, height = 1200)
```

```
FeaturePlot(total, features = c("CSPG4", "MCAM", "MYH11", "RAMP2", "RNASE1", "ENG", "EGFL7"))
```

```
dev.off()
```

```
#tumor, kidney epithelial cell, cycling
```

```
png("epi.png", width = 1600, height = 1200)
```

```
FeaturePlot(total, features = c("PAX8", "SLC26A7", "EPCAM", "MUC1", "MKI67", "TOP2A"))
```

```
dev.off()
```

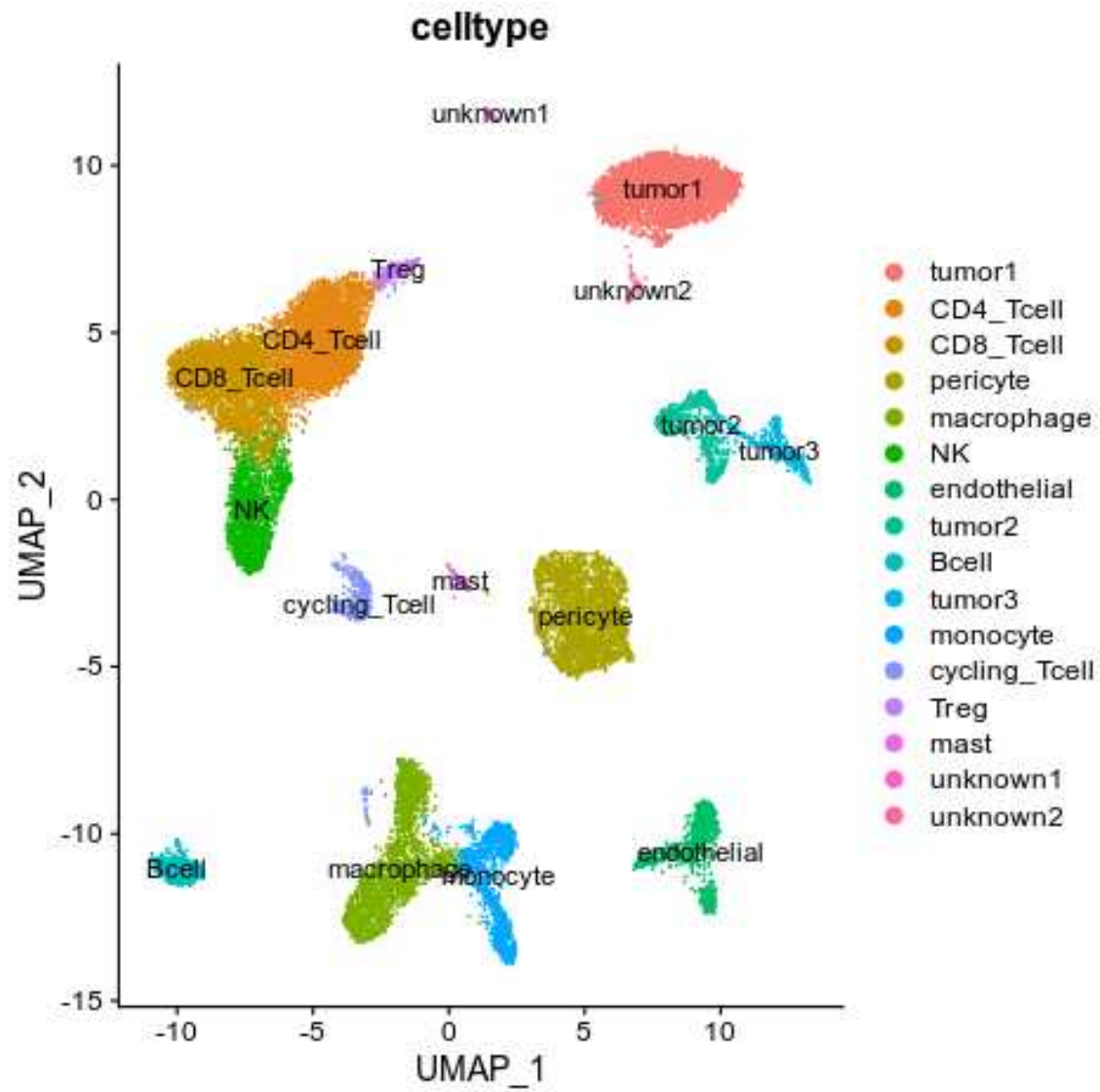
```
###
```

```
total <- Renameldents(total, '0' = 'tumor1', '1' = 'CD4_Tcell', '2' = 'CD8_Tcell', '3' = 'pericyte', '4' = 'macrophage', '5' = 'NK', '7' =  
'endothelial', '8' = 'tumor2', '9' = 'Bcell', '10' = 'tumor3', '6' = 'monocyte', '11' = 'cycling_Tcell', '12' = 'Treg', '13' = 'mast', '14' =  
'unknown1', '15' = 'unknown2')
```

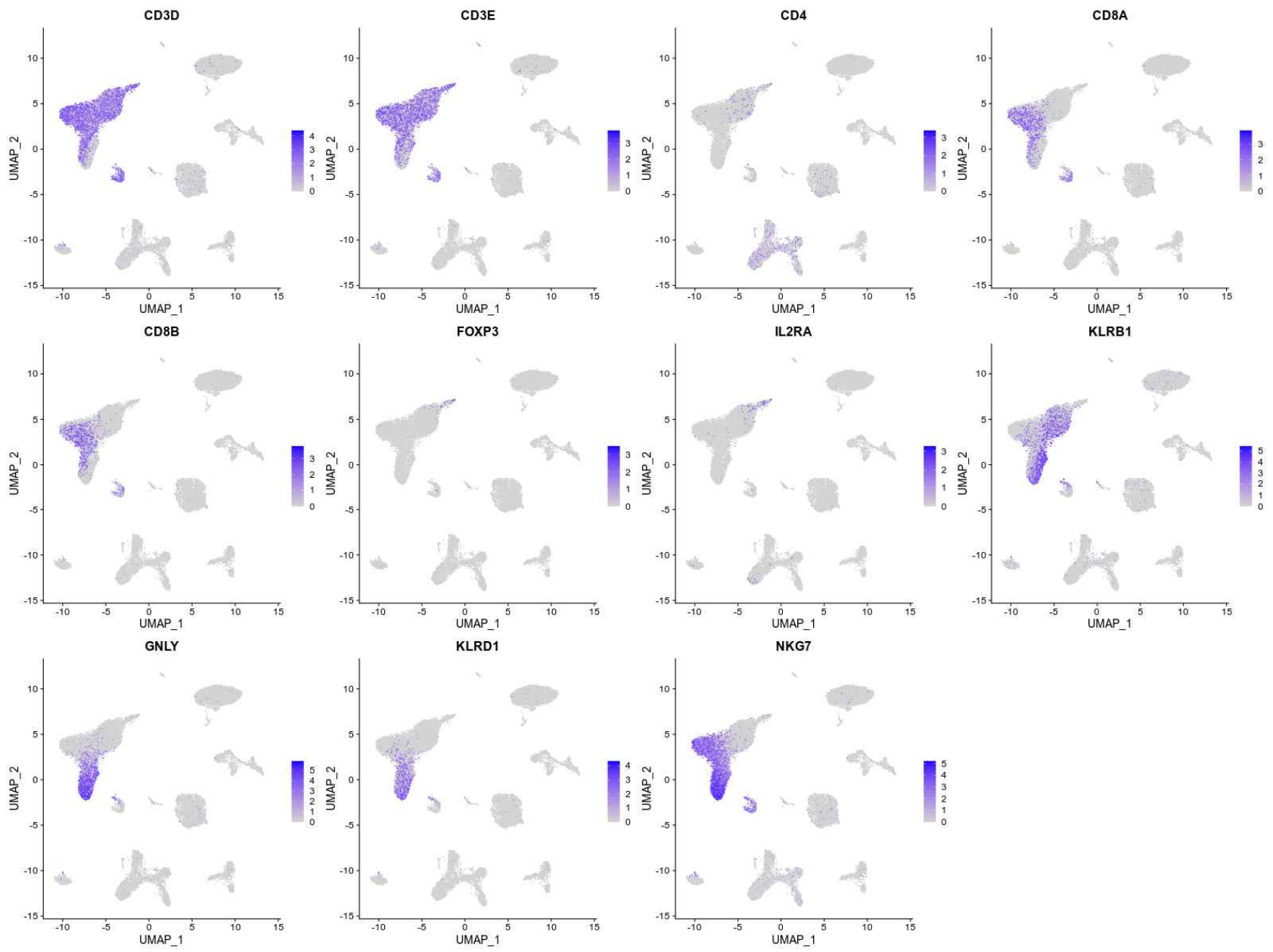
```
total$celltype <- total@active.ident
```

```
DimPlot(total, group.by = "celltype", label = T)
```

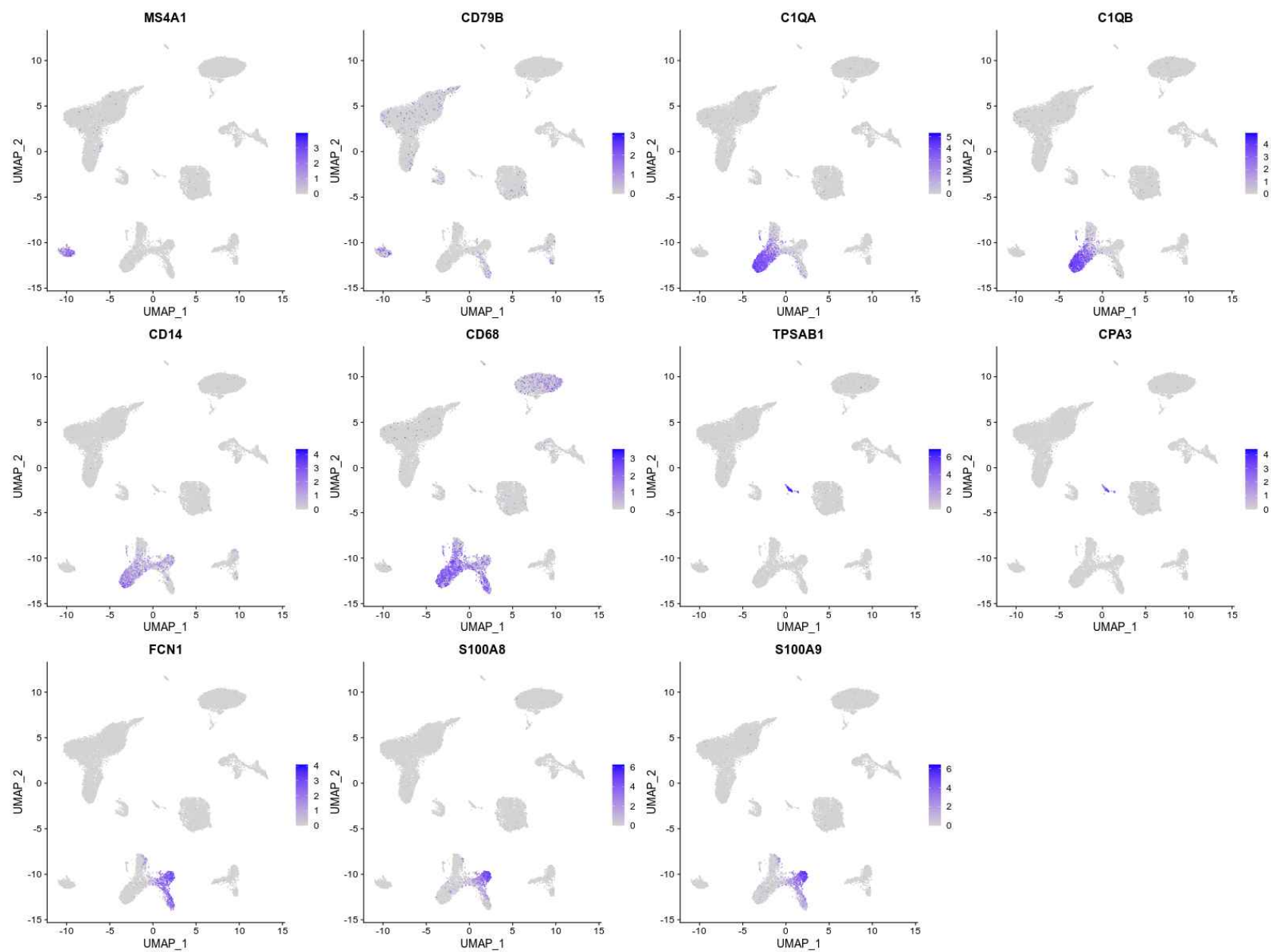
8) Celltype annotation



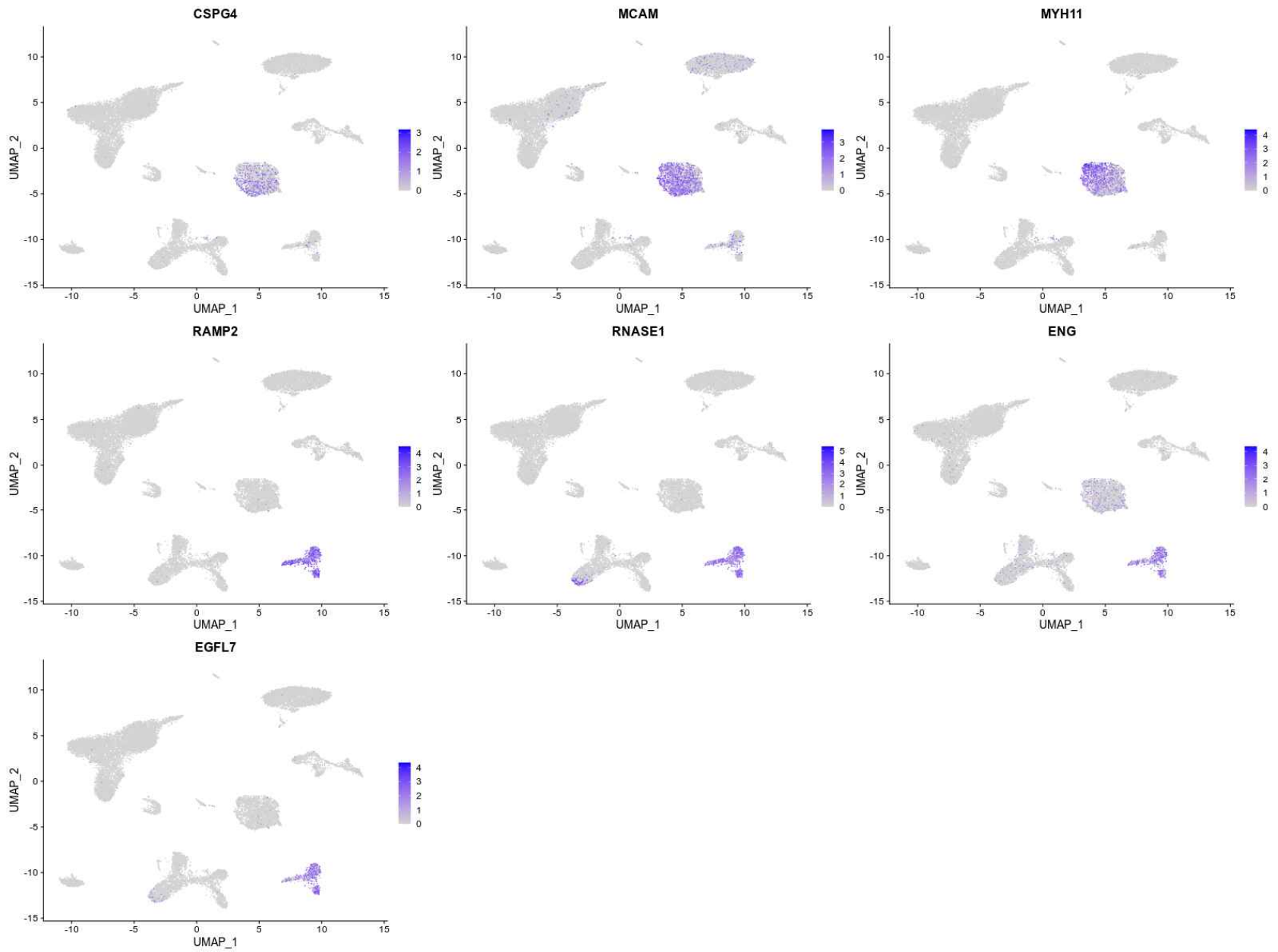
T cell and NK



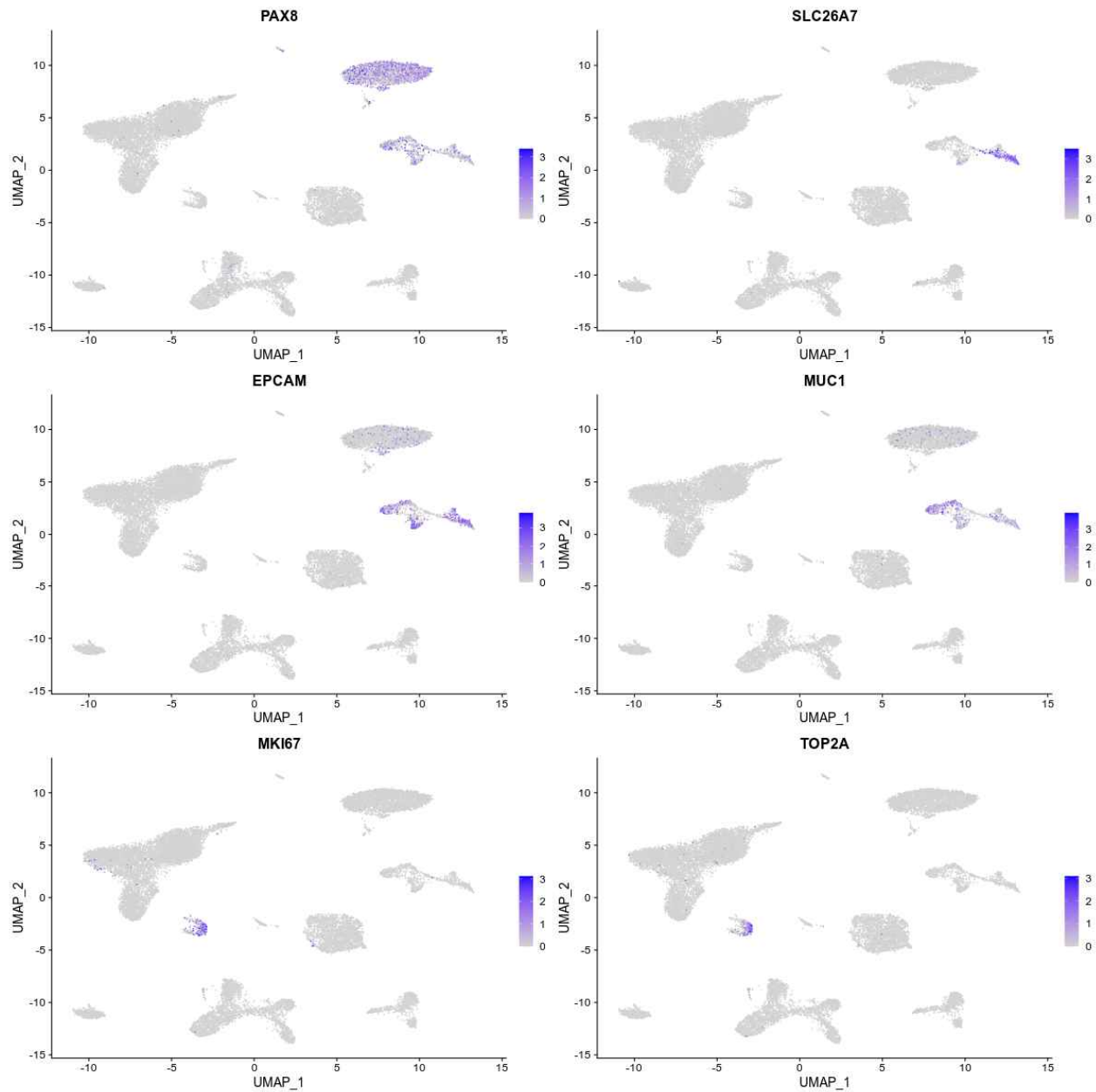
Bcell and myeloid



Stroma cell



Epithelial and tumor



9) Differentially expressed gene analysis between two groups

	p_val	avg_log2FC	pct.1	pct.2	p_val_adj
TMSB4X	1.180041e-123	0.8131724	1.000	1.000	3.863218e-119
CD74	1.336753e-110	2.4380974	0.841	0.460	4.376260e-106
HLA-DRA	6.658395e-84	2.6055562	0.616	0.121	2.179825e-79
RPS29	9.853846e-84	-0.6238238	0.997	1.000	3.225952e-79
RPS14	5.520321e-81	-0.6245685	0.993	1.000	1.807243e-76
RGS1	8.877664e-78	1.3781774	0.872	0.485	2.906370e-73
RPS27	2.936975e-74	-0.4452566	1.000	1.000	9.615067e-70
RPLP2	8.556250e-74	-0.5271463	0.998	1.000	2.801145e-69
DUSP4	1.420529e-72	2.1971333	0.553	0.075	4.650527e-68
RPL3	5.104196e-72	-0.6507585	0.982	0.992	1.671012e-67
EEF1A1	3.198898e-69	-0.5291642	0.997	1.000	1.047255e-64
RPS3	7.501267e-69	-0.5543631	0.994	0.998	2.455765e-64
HLA-DRB1	3.762663e-62	1.7883327	0.662	0.273	1.231821e-57
HLA-DPB1	1.903542e-61	1.7512534	0.649	0.275	6.231815e-57

GZMA: Tumor_CD8 T cell > Normal_CD8 T cell

→ Cytotoxic

HLA-DRA, HLA-DRB1, HLA-DPB1: Tumor_CD8 T cell >
Normal_CD8 T cell

→ activated

Script

```
total$group_celltype <- paste(total$group, total$celltype, sep = "_")
```

```
total@active.ident<-as.factor\(total\$group\_celltype\)
```

#Group: tumor or normal

#celltype: CD8_Tcell

```
cell1 <- "normal_CD8_Tcell"
```

```
cell2 <- "tumor_CD8_Tcell"
```

```
deg_cell <- FindMarkers(total, ident.1 = cell1, ident.2 = cell2, min.pct = 0.25, logfc.threshold = 0.25, only.pos = F)
```

avg_log2FC 양수가 ident.1 세포종이 과발현 하는 것

9) Differentially expressed gene analysis between two groups

Script

```
deg_cell <- FindMarkers(total, ident.1 = cell2, ident.2 = cell2, min.pct = 0.25, logfc.threshold = 0.25, only.pos = F)
```

deg_cell table 보는 방법

```
head(deg_cell)
```

#상단 5줄

```
tail(deg_cell)
```

#하단 5줄

몇줄만 보기

```
deg_cell[,c(1:10)]
```

→ 1줄 부터 10줄

```
deg_cell[,c(3:20)]
```

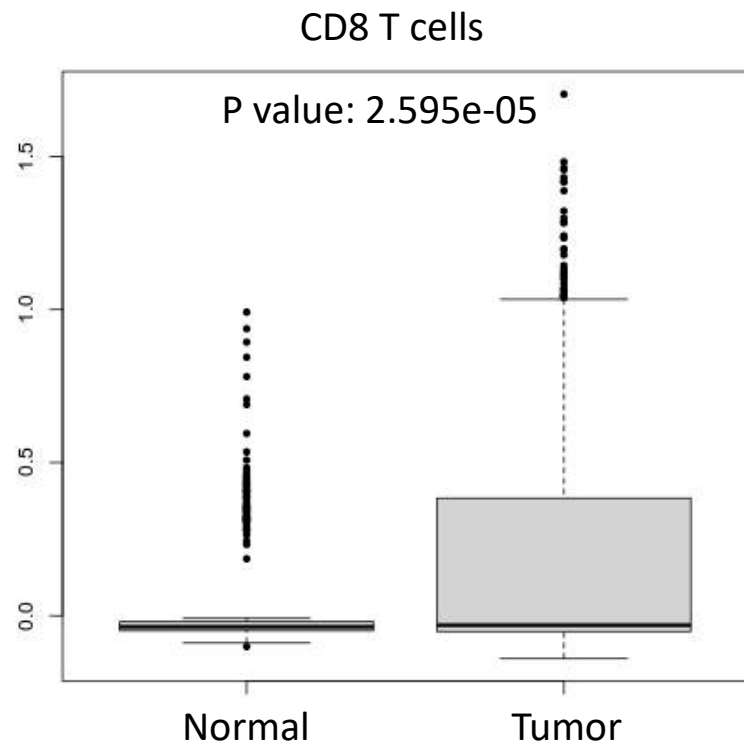
→ 3줄 부터 20줄

	p_val	avg_log2FC	pct.1	pct.2	p_val_adj
TMSB4X	1.180041e-123	0.8131724	1.000	1.000	3.863218e-119
CD74	1.336753e-110	2.4380974	0.841	0.460	4.376260e-106
HLA-DRA	6.658395e-84	2.6055562	0.616	0.121	2.179825e-79
RPS29	9.853846e-84	-0.6238238	0.997	1.000	3.225952e-79
RPS14	5.520321e-81	-0.6245685	0.993	1.000	1.807243e-76
RGS1	8.877664e-78	1.3781774	0.872	0.485	2.906370e-73
RPS27	2.936975e-74	-0.4452566	1.000	1.000	9.615067e-70
RPLP2	8.556250e-74	-0.5271463	0.998	1.000	2.801145e-69
DUSP4	1.420529e-72	2.1971333	0.553	0.075	4.650527e-68
RPL3	5.104196e-72	-0.6507585	0.982	0.992	1.671012e-67
EEF1A1	3.198898e-69	-0.5291642	0.997	1.000	1.047255e-64
RPS3	7.501267e-69	-0.5543631	0.994	0.998	2.455765e-64
HLA-DRB1	3.762663e-62	1.7883327	0.662	0.273	1.231821e-57
HLA-DPB1	1.903542e-61	1.7512534	0.649	0.275	6.231815e-57

10) Signature analysis

Tumor infiltrating T cell → might be exhausted

Exhaustion signature: PDCD1, CTLA4, HAVCR2, LAG3, TOX



→ T cells in the tumor-microenvironment are exhausted

Script

```
p<-c("PDCD1", "CTLA4", "LAG3", "HAVCR2", "TOX")
total<-AddModuleScore(total, features= p)
p1<- (total$Cluster1 +total$Cluster2+total$Cluster3+total$Cluster4+total$Cluster5)/5
total$exh <- p1
z<-subset(total, subset = celltype == "CD8_Tcell")
z1<-z$exh[which(z$group=="normal")]
z2<-z$exh[which(z$group=="tumor")]
wilcox.test(z1,z2)

boxplot(z1,z2, pch = 20, names = c("normal", "tumor"))
```

